Supplementary Material for "Drift Detection, Diagnosis and Reliability Analysis for Deployed Machine Learning Models"

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Section, figure, table and equation numbers without the prefix S refer to the main paper.

S1 Datasets

Each of the four datasets contains drift with a documented cause, so the verdicts of the pipeline can be checked against known events. Table S1 summarizes them. The two census tasks use data from California (CA), Texas (TX), New York (NY), Florida (FL) and Washington (WA).

Table S1 Datasets, drift windows and reference accuracy. Rows gives the smallest and largest window, Inputs the number of features or embedding dimensions and the last two columns the reference test accuracy of logistic regression (LR) and gradient boosting (GB)

						Ref. accuracy	
Dataset	Reference	Windows	Rows	Inputs	Classes	LR	GB
ACS Income	CA 2014	CA 2015–19 and 2021–23; TX, NY, FL, WA 2014	36,162–203,278	10	2	0.826	0.833
ACS Employment	CA 2014	CA 2015–19 and 2021–23; TX, NY, FL, WA 2014	56,576–326,241	16	2	0.729	0.772
Gas sensor	batches 1–2	batches 3–10	161–3,613	128	6	0.982	0.985
CLEAR-10	bucket 1	buckets 2–10	3,297–3,300	512	11	0.914	0.897

ACS Income and ACS Employment. Both tasks come from the American Community Survey (ACS) of the United States Census Bureau, as packaged by Folktables [1]. ACS Income predicts whether a person earns more than $50,000 from 10 features. ACS Employment predicts, from 16 features, whether a person aged 16 to 90 is employed. Nominal features, such as an occupation code, are treated as categorical because their numeric order carries no meaning. There is no standard 2020 release [2]. In 2019 the relationship variable RELP was replaced by RELSHIPP [3], which we map back to RELP using the code labels of the 2018 and 2019 data dictionaries. For GB, nominal codes beyond the 254 most frequent are pooled, which is the limit of the implementation.

Gas sensor array drift. This dataset from the University of California, Irvine (UCI) machine learning repository holds 13,910 measurements of six gases by 16 metal oxide sensors over 36 months, in ten batches [4, 5]. Each measurement has eight features per sensor. Batches 1 and 2 form the reference and batches 3 to 10 the windows. Because some batches are small, the number of audit labels for a window of n rows is given by $min(600, \lfloor n/2 \rfloor)$.

CLEAR-10. This image stream holds ten object classes and a background class in ten time-ordered buckets of about 3,300 images each [6]. Bucket 1 serves as the reference. Both models operate on the frozen ResNet-18 embeddings.

S2 Injection benchmark

The benchmark uses the 2014 data of 5 states (CA, TX, NY, FL, WA) for both ACS tasks. Each state is a reference distribution with its own model, split as in Section 3.2. For each of 10 seeds the split, the window and the injection are redrawn, which gives 50 replicates per configuration. Each replicate draws a production window of 4,000 rows and a grading holdout of 2,000 rows from the test part. Each mechanism is injected into both at three intensities.

Design A injects the four types in ways that partly match the tests. Covariate shift rescales two numeric features (ACS Income: AGEP and WKHP) for a random subset of rows. Prior shift drops rows of one class until the positive rate rises by 0.14, 0.23 and 0.32. Concept drift inverts the labels of the 15, 30 and 50% of rows with the highest value of one feature (ACS Income: SCHL). Label noise flips a random 8, 18 and 32% of labels. Each replicate also includes a control window without drift.

Design B uses mechanisms that the tests were not designed for. Covariate shift is selection bias: each row is kept with a probability given by $\sigma(\beta s(x))$, where σ is the logistic function, s is a seeded random score over all inputs and $\beta \in \{0.5, 1, 1.5\}$. The rows stay real and $P(Y \mid X)$ is exactly unchanged. Prior shift lowers the positive rate by 0.08, 0.14 and 0.20. Concept drift inverts the labels in the top 15, 30 and 50% of a second random score, which defines an

oblique region that no single feature describes. For ACS Income a second concept drift raises the income threshold to \$55,000, \$60,000 and \$70,000. Label noise is class-conditional, with positive labels flipped to negative at rates of 10, 20 and 35%.

The learned meta-classifier of Section 4.3 uses nine statistics of each window: the evidence $-\log_{10} p$ of each block test, the share of features with residual shift, the BBSE prior change, the CBPE change, the evidence and size of the audit accuracy drop and the evidence of the error structure test. Replicates are the unit of analysis in the paired tests because the windows of one replicate share a model and a reference.

S3 Methods and ablations

Table S2 repeats Table 2 of the paper with the three ablations of our procedure added.

Table S2 Share of benchmark windows with the correct drift type (%), including the ablations. Inc. and Emp. denote ACS Income and ACS Employment, A and B the two designs. An arrow marks a significant difference from ours after Holm correction: down means ours is better, up means ours is worse

	Logistic regression				Gradient boosting			
Method	Inc. A	Inc. B	Emp. A	Emp. B	Inc. A	Inc. B	Emp. A	Emp. B
Ours (hierarchical tests)	94.3	62.4	85.7	80.5	94.2	63.2	90.5	81.5
Input monitoring	30.3↓	20.0↓	30.5↓	25.0↓	30.3↓	20.0↓	30.5↓	25.0↓
BBSE first	30.5↓	27.7↓	30.5↓	39.2↓	30.5↓	28.1↓	30.6↓	40.3↓
Error-rate monitoring	52.3↓	48.5↓	52.2↓	62.8↓	52.5↓	49.6↓	53.1↓	64.8↓
Learned meta-classifier	97.5	31.1↓	95.8↑	48.2↓	98.5↑	39.2↓	97.4↑	51.2↓
Ours without MMD block	95.2	62.9	86.6	81.3	95.1	63.9	90.5	81.3
Ours, audit null without confidence	79.1↓	62.9	83.7↓	80.7	78.0↓	62.9	85.4↓	82.0
Ours, audit null with class cells	93.1	65.6↑	86.2	91.8↑	92.9↓	66.5↑	90.6	93.7↑

Removing the MMD block changes little on the benchmark, whose shifts are visible feature by feature. Removing the confidence term from the audit null lowers the accuracy on design A, because uniform label noise then looks like concept drift. Replacing the confidence bins by cells of predicted class and confidence helps on design B but not on ACS Income design A.

S4 Held-out mechanisms

Table S3 breaks the design B results down by mechanism.

Table S3 Design B results by drift mechanism. Corr. is the share of windows typed correctly (%) and Drop the mean true accuracy drop in points, where a negative value means that accuracy rose. The income threshold mechanism exists only for ACS Income

	ACS Income				ACS Employment			
	LR		GB		LR		GB	
Mechanism	Corr.	Drop	Corr.	Drop	Corr.	Drop	Corr.	Drop
Covariate: selection bias	99	0.2	99	0.3	100	−0.3	100	−0.3
Prior: decrease	92	−2.8	93	−2.4	97	2.5	97	1.9
Concept: oblique region	95	21.9	95	22.5	91	15.7	97	17.7
Concept: income threshold	4	−0.3	6	0.2	–	–	–	–
Noise: class-conditional	21	2.8	23	3.1	35	7.7	33	8.0

Three mechanisms transfer well: selection bias, a lower positive rate and concept drift in an oblique region. Raising the income threshold relabels some rows in every window but barely changes accuracy. Because the audit first asks whether accuracy fell, it reports no drift in most of these windows; the procedure is designed to find drift that harms the model, which this drift does not. Class-conditional label noise is the hardest mechanism. This is a genuine limitation of a null model that conditions on confidence alone.

The audit size controls how well the audit stage separates concept drift from label noise. Fig. S1 shows the share of correctly diagnosed windows on ACS Income design A with LR as the number of audit labels grows.

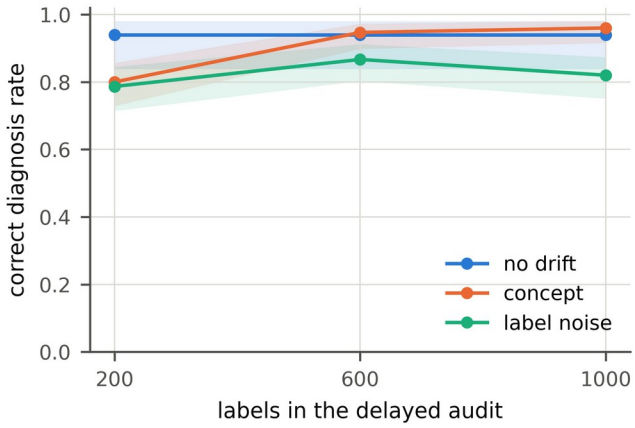


Fig. S1 Share of correctly diagnosed windows on ACS Income design A with LR against the number of audit labels. The shaded bands are 95% Wilson intervals

Concept drift is identified in 80% of windows with 200 labels and 95% with 600, while label noise is identified in 79% and 87%. The 600 labels used throughout the paper thus lie where the gain from more labels has largely flattened.

S5 Detection and accuracy estimation

Table S4 gives the alarm rate of each detector by injected type, pooled over the four design A configurations.

Table S4 Detector alarm rates on design A by injected type. The no drift column is the false alarm rate with its 95% confidence interval (CI)

	Alarm rate (%) by injected type				
Detector	No drift [95% CI]	Covariate	Prior	Concept	Noise
KS	1.0 [0.3, 3.6]	100.0	100.0	1.0	1.0
PSI	0.0 [0.0, 1.9]	99.7	0.3	0.0	0.0
MMD	7.0 [4.2, 11.4]	100.0	100.0	7.0	7.0
Confidence	5.0 [2.7, 9.0]	93.8	86.7	5.0	5.0
BBSD	2.5 [1.1, 5.7]	100.0	100.0	2.5	2.5
Domain classifier	6.0 [3.5, 10.2]	100.0	72.0	6.0	6.0

Over 200 control windows every interval contains $\alpha = 0.05$ or lies below it. Every detector catches covariate shift and all except PSI also catch most prior shifts. Concept drift and label noise windows share their inputs with the control window of their replicate, so no detector can see them, which is why the audit is needed.

The accuracy estimates are compared by their mean absolute error. On design A the errors of CBPE, AC, DoC and ATC stay below 0.83 points without drift and below 10.31 under covariate shift. Under concept drift and label noise every error equals the whole loss, about 14.71 points, because the inputs do not change. On the gas sensor data all four are off by 26.95 to 30.98 points on average. In batch 8 the LR model's mean confidence falls from 0.973 to 0.595 and its accuracy to 0.330, yet CBPE estimates 0.618.

S6 Real drift

Table S5 gives the details behind the ACS Income panel of Fig. 4.

Table S5 Detection and diagnosis on the ACS Income windows. Δ is the accuracy change in points and Est. err. the error of the CBPE estimate. A tick marks an alarm; Conf. denotes the confidence test, DC the domain classifier and Cov. a covariate verdict. Top shifts lists the two features with the largest effect size: occupation (OCCP), place of birth (POBP), relationship (RELP) and race (RAC1P)

	Accuracy				Alarms (LR)						Type	
Window	LR	ΔLR	ΔGB	Est. err.	KS	MMD	Conf.	BBSD	DC	Top shifts	LR	GB
CA 2014 (control)	0.826	0.0	0.0	0.0	–	–	–	–	–	–	–	–
CA 2015	0.824	−0.2	−0.2	+0.3	√	–	–	–	–	OCCP	–	–
CA 2016	0.821	−0.5	−0.5	+0.6	√	–	–	√	√	OCCP, POBP	Cov.	Cov.
CA 2017	0.817	−0.9	−0.9	+0.7	√	√	√	√	–	OCCP, POBP	Cov.	Cov.
CA 2018	0.795	−3.2	−2.2	+1.2	√	√	√	√	√	OCCP, POBP	Cov.	Cov.
CA 2019	0.784	−4.2	−3.0	+2.0	√	√	√	√	√	OCCP, RELP	Cov.	Cov.
CA 2021	0.770	−5.6	−4.4	+3.6	√	√	√	√	√	OCCP, RAC1P	Cov.	Cov.
CA 2022	0.760	−6.6	−5.1	+4.6	√	√	√	√	√	OCCP, RAC1P	Cov.	Cov.
CA 2023	0.749	−7.7	−6.1	+5.7	√	√	√	√	√	OCCP, RAC1P	Cov.	Cov.
TX 2014	0.802	−2.4	−2.7	+1.1	√	√	√	√	√	POBP, RAC1P	Cov.	Cov.
NY 2014	0.804	−2.2	−2.7	+0.8	√	√	√	√	√	POBP, RAC1P	Cov.	Cov.
FL 2014	0.786	−4.0	−4.3	+2.2	√	√	√	√	√	POBP, RAC1P	Cov.	Cov.
WA 2014	0.801	−2.5	−2.5	+0.9	√	√	√	√	√	POBP, RAC1P	Cov.	Cov.

The control window raises no alarm with either model. CA 2015, whose accuracy did not change significantly, is diagnosed as no drift by both models, while every other window is diagnosed as covariate shift. When the occupation codes were revised in 2018 [7], the model's reliance on occupation fell (SHAP change −0.032). From 2021 onwards race shifts, following the change in how the Census Bureau codes written race answers [8].

Every gas sensor batch is diagnosed as covariate shift. BBSE first labels these batches as prior shift because the predicted labels move, but the consistency test shows that they move because the sensors read differently, not because the mix of gases changed. On ACS Employment the inputs move, including relationship in 2019 and race from 2021, but the model is barely affected. On CLEAR-10 both models report no drift in buckets 2 and 3 and covariate shift in buckets 4, 5, 6, 7, 8, 9 and 10. In buckets 4, 5, 6 and 9 no single embedding dimension shifts significantly with LR. Only the MMD test detects the change, while the predicted labels stay put.

For the reliability layer, the gas sensor calibration part contains too few wrong predictions (4 with LR and 11 with GB), so the risk falls back to confidence. On the benchmark the fused risk changes AUROC by −0.002 to +0.009 relative to confidence alone.

S7 Protocol and configuration

All settings were fixed before the evaluation that uses them. The diagnosis and design B were both fixed before design B was first run. A later code review uncovered two errors in the diagnosis. The label-free stage had used a gatekeeping rule whose family-wise error rate reached 6.6% at a nominal 5% in a simulation, compared with 4.9% for the weighted Holm procedure of Section 3.3. The audit had also counted nominal categories over the reference and audit rows together. Both were corrected without regard to the results and every result was regenerated, which changed the verdict in 115 of the 5,300 benchmark windows.

Every permutation p value is given by $(1+b)/(1+B)$, where b of the B permutations reach the observed statistic [9]. The confidence test pools sparse histogram cells until every expected count reaches 5 [10]. The weighted feature tests of the label-free stage use Kish's effective sample size [11]. The risk fusion of the reliability layer is fitted only when the calibration part holds at least ten wrong predictions per signal [12]; otherwise the risk is simply $1-c$. Proportions carry Wilson intervals [13] and means carry bootstrap intervals over replicates. Permutation tests use 200 permutations, SHAP explains 1,000 rows per side against 100 background rows and each real audit uses 600 labels from its own random stream. SHAP is reported for LR only, because for GB the exact explainer would take hours per window. In the audit, a nominal category receives its own bin only if it occurs at least n_{min} times, where n_{min} is given by $\lceil 5/min(e, 1-e) \rceil$ and e is the audit error rate; the remaining categories share one bin. The density ratio weights of the triage are fitted on separate

folds and truncated at 20 times their mean. The importance-weighted holdout is 30% of the reference. One workstation with a 20-core processor and a graphics processor ran everything; the graphics processor computed the embeddings, permutation tests and large audit fits in double precision, with p values identical to the processor code. The software was scikit-learn [14], SciPy, NumPy, SHAP, OpenCLIP and PyTorch.

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